(MATH200) Differential Equations

Mathematical Concepts

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0 Notation and Conventions

General mathematical notation and writing follow *(MATH000) Mathematical Language*. The conventions below record only notation specific to differential equations used in this book. When several equivalent notations are listed, the first form is used throughout.

Differential Equations

The Fourier transform uses the normalization

$$\widehat{f}(\boldsymbol{\xi}) := \int_{\mathbb{R}^n} f(\mathbf{x}) e^{-2\pi i \mathbf{x}\cdot\boldsymbol{\xi}}\, \mathrm{d}\mathbf{x}, \qquad f(\mathbf{x}) = \int_{\mathbb{R}^n} \widehat{f}(\boldsymbol{\xi}) e^{2\pi i \mathbf{x}\cdot\boldsymbol{\xi}}\, \mathrm{d}\boldsymbol{\xi},$$

whenever the Fourier inversion formula applies.

Notation	Meaning
y'	first derivative with respect to the understood independent variable
y''	second derivative
$\dot{x}$	first derivative of x with respect to time
$\ddot{x}$	second derivative of x with respect to time
$y_h,\ y_c$	homogeneous or complementary part of a solution
y_p	particular solution
$c_1, c_2, \ldots$	arbitrary constants in a general solution
$W(y_1, y_2)(t),\ W[y_1, y_2](t)$	Wronskian of y_1 and y_2
$\mathcal{L}\{f(t)\}(s)$	Laplace transform of f
$\widehat{f}(\boldsymbol{\xi}),\ \mathcal{F}f(\boldsymbol{\xi})$	Fourier transform under the normalization above

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